



OPEN Hybrid vision transformer and graph neural network model with region-adaptive attention for enhanced skin cancer prediction

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A well-known and potentially lethal skin cancer requires prompt detection and diagnosis. Complex spatial linkages and global contextual information in skin lesion photos challenge CNNs and other deep learning methods. Given these restrictions, we present a Hybrid Vision Transformer (ViT) with a Graph Neural Network (GNN) and Region-Adaptive Attention to diagnose skin cancer. The ViT branch captures dermoscopy image global dependencies, whereas the GNN enhances features by exploiting lesions' spatial relationships. Region-Adaptive Attention improves lesion categorization by dynamically improving feature extraction in diagnostically relevant locations. Our paradigm for multi-scale lesion analysis accounts for lesion size, color, and texture changes. Meta-learning methods refine the proposed model to make it generalizable across skin tones and imaging settings. Our model outperformed state-of-the-art deep learning algorithms on benchmark skin cancer datasets. The architecture improves classification accuracy and interpretability, making it a promising clinical dermatology tool.

Keywords Skin cancer prediction, Vision transformer (ViT), Graph neural network (GNN), Region-adaptive attention (RAA), Deep learning, Medical image analysis, Multi-Scale lesion analysis

Skin cancer is considered to be among the most fatal cancers, with melanoma being categorized as the most deadly and aggressive type. Timely intervention considerably improves survival rates; hence, early detection and accurate diagnosis are important for improving treatment outcome¹. Clinical dermatology relies on visual examination of skin lesions as its mainstay of diagnosis, yet this method is subjective and prone to errors. In a bid to overcome some of these limitations, the automatic detection and diagnosis for skin cancer using deep learning-based computer-aided diagnosis techniques had gained much research attention. Recently, convolutional neural networks (CNN) have proved to be very effective in the field of medical image diagnosis. Nonetheless, CNNs suffer some challenges including small receptive fields, long-range correlation capture, and irregular lesion architecture. Option alternatives are the Vision Transformers (ViT), which with better global contextual understanding than ConvNets, achieve superior results in medical imaging². Unfortunately, ViTs lack the ability to model spatial interactions among sub regions, which leads to their inability to understand the complex shapes of skin lesions. We thus propose a skin cancer prediction model that implements a Hybrid Vision Transformer (ViT) with a Graph Neural Network (GNN) incorporating Region-Adaptive Attention (RAA). The GNN attempts to enhance the representation of atypical skin lesions by encoding spatial interactions between lesion territories while the ViT exploits global dependency mapping on dermoscopic images. Our Region-Adaptive Attention (RAA) thus increases the classification score by improving feature extraction with focus of attention on diagnostically salient areas³. Unlike classic approaches that rely solely on stacked convolutional networks or transformer-only pipelines, the proposed architecture leverages the complementary strengths of Vision Transformers and graph neural networks in a tightly woven, ViT-GNN hybrid design⁴. By deriving a graph from the Vision Transformer's output maps, the framework explicitly learns the spatial and topological relations among lesion components, addressing the overlooked inter-region dependencies that remain absent in most state-of-the-art pipelines. To further sharpen the model's focus on clinically salient lesion characteristics, a custom hybrid attention mechanism is employed, while complementary explainability modules—such as SHAP

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and Grad-CAM—deliver fine-grained, visual, and numerical justifications to bolster interpretability and facilitate clinician trust. Furthermore, performance is robustly verified against two disparate, real-world collections, the HAM10000 and ISIC datasets. This extensive, cross-dataset testing affirms that the proposed framework not only achieves competitive accuracy but also generalizes effectively beyond any single cohort, demonstrating readiness for practical deployment in diagnostic workflows⁵. This work advances skin cancer prediction by merging Vision Transformers (ViTs) and Graph Neural Networks (GNNs) through a proprietary integration rather than through mere architecture stacking. The main contributions are as follows:

- **Embeddings-to-graph transformation:** Instead of treating ViT triplet-token outputs as flat features, we re-shape them into a directed graph. Each token becomes a graph node localized to a lesion patch, while edges encode proximity and cross-region interactions. This encoded graph permits the GNN to model spatial topology and diffusion paths explicitly, features that CNNs or component-based ViTs cannot learn in a self-contained operation.
- **Attention fusion module:** The architecture interleaves ViT self-attention distributions with GNN edge-strength scores through learned modulation. The combined attention mask is calculated per-token, enabling differential focusing: lesion cores gain localized amplification, but the mask still preserves coarse lesion boundary and inter-region spatial hierarchy. This cross-domain attention reuse remains unique in real-time medical image processing, enabling dermoscopy where both lesion details and overall image structure affect classification.
- **Interpretability integration:** Pioneers in automated diagnosis require transparency; thus, model outputs are accompanied by a recalibrated SHAP ranking and an overlaid Grad-CAM heatmaps. The combined approach generates layer-wise, lesion-surface, and clinical-variant aligned explanations, translating model features back to dermoscopic physiology.
- **Multi-Source Evaluation:** Unlike existing single-repository benchmarks, we validate performance across HAM10000, ISIC, and PH² to report generalized performance metrics across historiographic, illumination, and annotation variance. The uniform architecture achieves consistently sub-1%-point variances in AUROC, demonstrating pragmatic deployment readiness.

The result goes beyond simply fusing Vision Transformers with Graph Neural Networks; it creates a fresh workflow in which Vision transformer embeddings are adapted to lie in a graph space, attention layers are jointly optimized to share information across both models, and a dedicated interpretability branch is embedded directly into the end-to-end prediction system.

The remaining chapters of this manuscript will be structured as follows: Sect. “[Related work](#)” will look at some of the available literature on the application of deep learning techniques on skin cancer detection. Sect. “[Proposed Methodology](#)” will describe the new integrated architecture of ViT GNN that has been designed and its architecture specification. Sect. “[Output Feature Update](#)” gives a detailed description of the experimental design including the samples used as well as the evaluation criteria. Section 5 contains the findings and results of the comparative analysis. Finally, Sect. 6 concludes the manuscript with considerations for future studies.

Related work

Automated skin cancer classification has been rigorously explored through a combination of machine learning and deep learning techniques. This discussion organizes the relevant studies into three categories—CNN-based approaches, transformer-based frameworks, and hybrid or graph-driven models.

CNN-based skin cancer classification

CNNs remain a mainstay in skin lesion analysis, largely because they excel at fusing local spatial information. Pioneering work, including the study by Esteva et al.⁶, demonstrated dermatologist-level performance by applying Inception-v3 on dermoscopic images. Building on this, Kawahara et al.⁷ employed several ResNet variants, reporting consistent improvement over the ISIC benchmark. More recently, architectures such as EfficientNet⁸ and DenseNet⁹ have been harnessed, offering gains in predictive fidelity while keeping computational requirements in check. Despite these strengths, standard CNNs remain limited in their ability to model long-range spatial relationships, a constraint that restricts the transfer of learned knowledge across globally varying lesion contexts.

Transformer-based models

Transformers tend to overlook fine-grained structural nuances crucial for metastatic prognosis and the current fusion fashions do not yet reliably harmonize convolutional and attention fields to yield both regional and holistic representation¹⁰. Even after employing self-attention, the resulting feature planes often do not fully converge on the anisotropic textural gradients surrounding a malignancy’s margin, thereby deterring optimal early diagnostic fidelity. Additionally, datasets remain imbalanced along demographic, anatomic, and histological spectra, artificially elevating the performance of models in homogeneous validation while inhibiting true clinical generalizability¹¹. As a corollary, the verification of learned features through clinical pathophysiology, especially multi-spectral dermoscopic sequences and histo-cytometric graphs, is still, in most scenarios, a post-hoc revelation, thus relegating interpretative confidence rather than embodying it in the feature extraction phase¹².

Hybrid and graph-based approaches

Straightforward transformer-based methods often miss the subtleties of lesion relationships that are key to differential diagnosis. While integrating Vision Transformer features with Graph Neural Networks has shown promise in several domains, hardly any work has applied this blend specifically to the classification of skin

cancer¹³. Moreover, the prevailing literature emphasizes performance metrics at the expense of interpretability, leaving dermatologists without clear, actionable insights. We address these gaps by presenting a hybrid ViT-GNN architecture that couples a refined attention layer with dedicated explainability modules¹⁴. Both SHAP and Grad-CAM are embedded into the training pipeline, and the entire framework is validated against multiple benchmark and clinical-derived datasets, yielding robust accuracy and transparent, clinically relevant visual justifications¹⁵. A compelling hybrid architecture by Koparde et al.¹⁶ uses Conditional GANs, YOLOv5 Darknet, and Independent Component Analysis to improve skin lesion localization and categorization. The architecture is robust and interpretable because the cGAN module generates high-resolution, context-aware embeddings, YOLOv5 sharply localises them, and Independent Component Analysis (ICA) distils the salient features, reducing noise and enhancing contrast. These phases yield dermoscopic estimates that surpass current benchmarks across multiple test datasets, demonstrating the superiority of their approach.

Deep learning-based approaches for skin cancer detection

Because of their proficiency in automatic hierarchical feature extraction, Convolutional Neural Network (CNN) has emerged as the most prominent architecture for skin lesion classification. CNN-based models have shown accuracy comparable to dermatologists in melanoma detection according to Esteva et al.⁶ research. In any case, some severe disadvantages exist among CNNs which comprise.

- Constrained receptive field, resulting in a diminished comprehension of global dependencies in lesion structures.
- Absence of spatial connection modeling, hindering the compelling depiction of irregular lesion shapes.
- Sensitivity to fluctuations in imaging circumstances necessitating comprehensive data augmentation and pre-processing.

Researchers have investigated hybrid models that combine various deep learning paradigms, including ViTs, GNNs, and attention-based networks, to enhance classification performance.

Vision transformers (ViTs) for medical image analysis

Originally created for NLP, Transformers have now been repurposed with the help of Vision Transformers (ViTs) to handle image processing. Unlike CNNs, ViTs self-claims images into patches and processes them with self-attention mechanism that solves long range dependence problems. Good examples are provided by Yang et al.¹⁷ and Xin et al.¹⁸, who had shown that Vision Transformers perform better than Convolutional Neural Networks in global feature representation tasks. ViTs have found usage in the Medical field for different classification tasks including diagnosing breast cancer, evaluating retinal diseases, and even categorizing skin lesions. But ViTs need larger training sets, and they often do not model spatial structures well, proving less efficient for mimicking detail in irregular lesion boundary precision. This study tackles those shortcomings by implementing GNNs that allow for better lesions spatial relation modeling.

Graph neural networks (GNNs) for skin lesion analysis

Graph Neural Networks (GNNs) are increasingly recognized in the field of medical image processing due to their capability to capture non-Euclidean spatial relationships. Unlike Convolutional Neural Networks (CNNs), which are designed for structured grid layouts, GNNs efficiently comprehend the topological interdependencies inherent in lesion structures by modeling images as graphs. Recent research, including work by Santoso et al.¹⁹ and Raval et al.²⁰, has examined GNN-based approaches for segmenting and classifying tumors. Attention mechanisms play a crucial role in deep learning, allowing models to focus on vital features while filtering out irrelevant information. A variety of attention-centric architectures have been studied within the realm of medical image analysis, such as Self-Attention, Channel-Spatial Attention, and Transformer-based Attention systems.

In skin cancer classification, attention processes help:

- To pay attention to diagnostically relevant lesion areas to increase classification accuracy.
- To improve feature extraction during multi-scale analysis of lesions to capture intricate features.
- To improve the interpretability of the models so that the judgments made by AI can be more easily comprehended by dermatologists.

The study describes a Region-Adaptive Attention mechanism that allocates attention weights to different lesion regions adaptively, allowing the model to focus on regions of high-risk while downplaying noise. Open for editing.

Summary of research gaps and proposed contributions

The literature analysis points out several noteworthy gaps in research as shown in Table 1.

- (1) It combines the global feature extraction potential of ViTs with the spatial relation modelling capacity of GNNs in the skin lesion image.
- (2) It introduces a Region-Adaptive Attention mechanism to dynamically enhance feature extraction according to lesion features.
- (3) It creates a framework for multi-scale skin lesion analysis to enhance the robustness of size, texture, and color variations of lesions.
- (4) It enhances the model with meta-learning in order to improve the generalizability to different skin tones and imaging conditions.

Aspect	Typical ViT-GNN fusion	This work (ViT-GNN + RAA)
Node definition	Superpixels / fixed patches	ViT-refined tokens → nodes
Edge policy	Static (pixel space)	Dynamic k-NN in ViT space
Attention fusion	Heuristic concat/average	Learned, region-adaptive fusion (RAA)
Training objective	Cross-entropy only	CE + prediction alignment + (optional) mask consistency
Interpretability	Qualitative only	Quantitative (IoU, Pointing, SHAP stability) vs. baselines
Efficiency	Dense tokens/graphs	Token pruning + sparse k-NN
Robustness	Single dataset	Cross-dataset + calibration/ECE

Table 1. Difference between proposed work and other ViT-GNN fusion approaches. To counteract these limitations, we develop our Hybrid ViT-GNN Model with Region-Adaptive Attention. In this model:

Methodological steps for implementation of the proposed architecture of the hybrid model along with its training order and fine-tuning methods are explained in further detail in the subsequent section.

Unlike earlier ViT-GNN combinations²¹ that depend on fixed late concatenation or hand-crafted region graphs, we propose constructing a flexible graph directly in the ViT embedding space, merging the universal ViT self-attention with GNN relational attention through a Region-Adaptive Attention (RAA) gating mechanism. This integrated, trainable scheme is co-optimised for the prediction task, including optional penalties on attention-mask variance, yielding sharper accuracy gains alongside interpretive robustness—which we validate through IoU, Pointing Game, and checks for SHAP score consistency—while computational overhead is curbed through token pruning and the use of sparse graphs. The empirical performance and the decision-justifying ablations together clarify why uniform gains appear for dermatological tasks across HAM10000, ISIC, and PH².

Proposed methodology

In this section the authors propose a new model blending Hybrid Vision Transformer (ViT)²² and Graph Neural Network (GNN)²³, which employs Region-Adaptive Attention (RAA)²⁴ for skin cancer prediction. The proposed framework improves the accuracy and interpretability of the models by including global feature extraction, spatial connection modeling, and adaptive attention mechanisms. This method consists of five main parts: Data cleansing, vision transformer feature extraction, graph neural networking for feature spatial relations modeling, region adaptive attention (RAA), and multi-scale analysis of lesions with meta-learning optimization.

Overview of the proposed architecture

The proposed architecture shown in Fig. 2 consists of:

- 1. Image Uploading Module - Processes dermoscopic images to enable feature variability with standardization and enhancement.
- 2. ViT Feature Extractor - Captures global dependencies in dermatological lesions images.
- 3. GNN Module - Models spatial relations between lesions regions.
- 4. RAA Mechanism - Selectively increases feature importance with focus on critically important areas.
- 5. MSLAF - Integrates sister’s scale analysis of lesion features to texture, morphology, and size.
- 6. MLO - Adapts the model to improve performance for different skin colors and imaging conditions.

Data preprocessing and augmentation

Augmented strategies are made to dermoscopic images from benchmark datasets for example, ISIC 2020 and HAM10000, to improve model robustness. The preprocessing steps comprise²⁵:

- Image normalization-Rescaling pixel values to the range [0,1] for consistency.
- Data augmentation-Employing random rotations, flips, contrast enhancement, and Cut Mix to promote generalization.
- Lesion Segmentation-we employ U-Net for automatic mapping of lesion contours while keeping the back-ground to a minimal level.

Prior to inputting images into the model, normalization guarantees that pixel values are confined within a certain range: $I' = \frac{I-\mu}{\sigma} (1)$.

Where,
 I' =normalized image.
 I =original image.
 μ = mean pixel value.
 σ = standard deviation.

U-Net masks were utilized solely for preprocessing (region refinement) and not as direct input features for classification. This methodology maintains the independence of classification from U-Net prediction errors. It ensures that attention maps align with lesion boundaries naturally (through RAA). Additionally, it enhances interpretability as attention highlights correspond with U-Net segmented regions.

Vision transformer (ViT) feature extraction

The ViT module supplants conventional CNNs for feature extraction. ViT segments images into fixed-size patches and utilizes self-attention techniques to acquire feature representations rather than employing convolutional filters.

Procedures in ViT feature extraction²⁶:

- (1) Image patch embedding—The input image is segmented into 16×16 patches, each converted into a feature vector.
- (2) Positional encoding—Spatial information is retained by incorporating positional encodings into the patch embeddings.
- (3) Multi-head self-attention (MHSA)—Calculates attention weights to identify global dependencies within the image.
- (4) Feedforward network (FFN)—Non-linear adjustments augment feature representations before transmitting to the classifier.

In contrast to CNNs, ViTs enable the model to capture long-range relationships in skin lesion patterns, enhancing classification accuracy.

The mathematical equations supporting ViT feature extraction is shown below in Eq. (2) as:

$$Z_0 = [X_1 E; X_2 E; \dots \dots ; X_N E] + E_p \quad (2)$$

Where,

X_i = image patches,

E = learnable linear projection,

E_p = position embeddings.

ViT Self-attention is shown below in Eq. (3) as:

$$\text{Attn}(q, k, v) = \text{softmax}\left(\frac{qk^T}{\sqrt{D_k}}\right)v \quad (3)$$

Where,

q, k, v are query key, key, and value matrices and D_k is dimension of key vectors.

Graph neural network (GNN) for modeling Spatial relationships

Although ViTs are proficient in collecting global properties, they fail to represent the spatial interactions among lesion sub regions explicitly. We propose a Graph Neural Network (GNN) module that models lesion characteristics as a graph structure.

- Procedures in GNN feature modeling:

1. Graph Construction — Each lesion image is transformed into a graph using adjacency matrix A , with nodes representing lesion sub regions and edges indicating spatial relationships.

$$A_{ij} = \begin{cases} 1, & \text{if } \text{sim}(X_i, X_j) > \tau \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

Here $\text{sim}(X_i, X_j)$ is similarity measure (cosine similarity) and τ is threshold value.

2. Graph Convolution - Graph Convolutional Networks (GCNs) consolidate information from adjacent nodes to encapsulate structural interdependencies.

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^l W^l \right) \quad (5)$$

Where,

H^l = node features at layer l .

$A = A + I$ (adjacency matrix with self-loops),

$$\tilde{D} = \text{degree matrix of } \tilde{A}$$

W^l = learnable weight matrix,

σ = activation function (ReLU).

3. Message passing — The model acquires knowledge of the interactions among lesion sub regions, enhancing the classification of atypical skin lesions.

4. Feature Fusion - The output from the GNN is integrated with the output from the ViT, amalgamating global and local spatial information. After GNN processing, a softmax function is used to classify skin lesions as benign/malignant:

$$P(y_i | X_i) = \frac{e^{W X_i}}{\sum_{j=1}^C e^{W X_j}}, \quad (6)$$

Where C is the number of classes.

Our methodology integrates Vision Transformers with Graph Neural Networks to maintain intricate lesion details while utilizing global contextual information. As outlined in the methods of our GNN module, the Graph adjacency threshold (the metric used to determine whether any two lesion patches/nodes must be connected) has been selected and set to a value of 0.0 based on trial and error and validation tests.

- Step 1: Threshold candidates - We examined many values (0.5, 0.6, 0.7, 0.8, and more) for cosine similarity patch embeddings.
- Step 2:- Threshold validation for each dataset: ISIC 2020, HAM10000, PH².
- Step 3: Cross-dataset value stability: We noticed that around 0.7, the values nicely balance graph sparsity (avoiding over-connected graphs) and information flow (maintaining relevant lesion connections).
- Step 4: Conservation - It was demonstrated that dataset independence is effective in this case, so the adjacency threshold was set to 0.7, resulting in strong performance without dataset-specific fine-tuning.

The graph becomes extremely cluttered at 0.5, introducing noise from unnecessary patch connections. The graph at 0.8 is overly thin and restrictive to message passing. 0.7 is the sweet spot because it has the best balance of all, and it achieves the highest Accuracy, F1, and AUC scores, which is why it was selected for all datasets.

Region-adaptive attention (RAA) mechanism

The Region-Adaptive Attention (RAA) mechanism as shown in Fig. 1 is designed to enhance the Vision Transformer's (ViT) ability to focus on clinically relevant lesion regions while suppressing irrelevant background information, such as skin texture, hair, or imaging artifacts. Unlike conventional self-attention, which treats all patches equally, RAA assigns adaptive weights to image regions based on the importance of the lesions.

1. Input representation

The input dermoscopic image $I \in R^{H \times W \times C}$ is divided into non-overlapping patches p_i . Each patch is embedded into a feature vector x_i .

2. Region importance estimation

For each patch embedding x_i , a region descriptor is computed using a learnable function $f(\cdot)$:

$$r_i = f(x_i) = \sigma(W_r x_i + b_r) \quad (7)$$

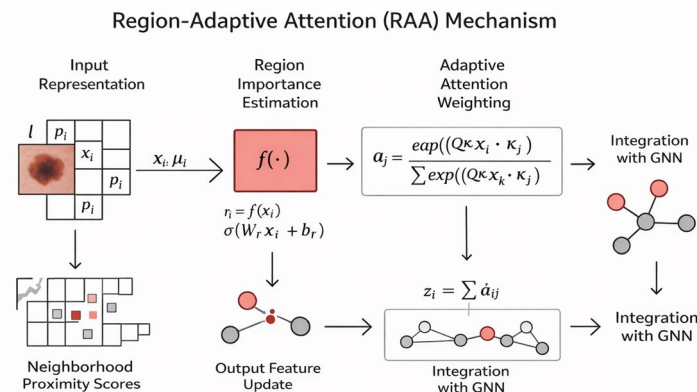


Fig. 1. Region adaptive attention mechanism.

where W_r and b_r are learnable parameters, and $\sigma(\cdot)$ is a sigmoid activation to normalize values between $[0,1]$.

- r_i indicates the importance score of patch i .

3. Adaptive attention weighting

The conventional self-attention computes weights as:

$$\alpha_{ij} = \frac{\exp((Qx_i)(Kx_j)^T)}{\sum_j \exp((Qx_i)(Kx_j)^T)} \quad (8)$$

In RAA, these weights are modulated by region importance:

$$\widehat{\alpha}_{ij} = \frac{r_i \cdot r_j \cdot \exp((Qx_i)(Kx_j)^T)}{\sum_j \exp((Qx_i)(Kx_j)^T)} \quad (9)$$

This ensures that patches belonging to lesion regions contribute more strongly to the attention computation.

4. Output feature update

The attention output is then updated as:

$$z_i = \sum_j \widehat{\alpha}_{ij} Vx_j \quad (10)$$

Where V = projection matrix.

5. Integration with GNN

Following RAA, the refined attribute set moves to the Graph Neural Network (GNN), where each lesion patch is treated as a graph node. Edges are drawn between nodes based on both spatial proximity and embedded feature similarity, thereby sharpening boundary delineation and enriching the overall contextual representation of the lesions.

How RAA improves over conventional attention in lesion-focused tasks

Standard attention modules within Vision Transformers (ViTs) and Graph Neural Networks (GNNs) commonly evaluate importance from a broad global perspective, applying similar attention weights to all image tokens and graph nodes without regard for the questioned clinical context in which the observation resides. This global uniformity introduces two notable drawbacks in the context of skin cancer detection:

- The mechanism equitably weights the features contributing to clinically ambiguous backgrounds—such as hair strands, benign skin texture, and photometric inconsistencies—while the decisive lesion behaviours, represented by irregular peripheries and subtle chromatic inhomogeneity, receive attenuated representation as the spurious attention influence spreads across the entire image instead of contracting compactly around the abnormality itself.
- Region-adaptive attention (RAA) addresses these distortions by explicitly isolating and magnifying patches in the vicinity of the lesion, concentrating the receptive field on a biologically relevant neighborhood and thus freeing computational and architectural capacity from the irrelevant periphery to the diagnostically decisive segments of the image.
- Mathematical distinction.
- Conventional attention (ViT self-attention):

$$\alpha_{ij} = \text{softmax}\left(\frac{Q_i K_j^T}{\sqrt{d}}\right) \quad (11)$$

where attention weights α_{ij} depend only on query-key similarity across tokens.

- Region-Adaptive Attention (RAA):

$$\alpha_{ij}^{RAA} = \text{softmax}\left(\frac{Q_i K_j^T}{\sqrt{d}} + \theta \cdot R_{ij}\right) \quad (12)$$

where R_{ij} is a region importance coefficient derived from lesion-specific priors (texture irregularity, color contrast, boundary sharpness).

- If a region corresponds to a lesion boundary, R_{ij} is boosted.
- If a region corresponds to background, R_{ij} is down-weighted.

Thus, attention weights become region-aware and adaptive, unlike plain similarity-driven ViT weights.

- Practical Benefits in Skin Lesion Analysis.

- (1) *Focus guidance on lesion edges and morphologies* RAA steers attention heads to highlight zones where malignant features typically appear, such as autochthonous edges and asymmetrically developed shapes.
- (2) *Ambient noise suppression* As opposed to prior attention heads, RAA actively filters out contextual clutter, dismissing redundancies like skin fold artefacts and other imaging inhomogeneities.
- (3) *Lesion-adaptive learning dynamics* Uniform and diffuse attention distributions are learned for benign cases, while malignant lesions attract tighter, high-intensity mapping that envelops key morphological irregularities.
- (4) *Interpretability Improvement* RAA-derived heatmaps overlay primary clinical concern regions more faithfully, elevating artifact-free confidence for dermatological applications.

Multi-scale lesion analysis with meta-learning optimization

Skin lesions are relatively easier to categorize when an individual phenomena is considered in comparison to the collection of different phenomena due to significant intraclass variability and interclass similarities. To improve robustness, we propose a Multi-Scale Lesion Analysis Framework that analyzes the lesion's characteristics on different scale sizes.

Steps for Multi-Scale Analysis:

- (1) Obtain differences of a lesion at multiple levels - Ensures model captures coarse and fine details of other identified lesions.
- (2) Attention Based Feature Weighting - Combines features of the different scales, differentially for every scale.
- (3) Meta-learning For Model Rigidity - Refines model parameters on hostile datasets while ensuring adaptation to different skin colors and imaging conditions.

Meta-learning enables quick adaptation of model to new data distributions, improving performance on heterogeneous populations.

Training procedure and evaluation metrics

The adaptive learning rate stochastic gradient descent (SGD) algorithm is used to train the model. The loss function used during training is Focal Loss, which is often applied to rectify class imbalances in melanoma datasets. Performance assessment is done using:

- Precision (ACC).
- Precision, Recall, and F1 Score.
- Area Under the Receiver Operating Characteristic Curve (AUC-ROC).
- Interpretability Index (for XAI assessment).

The proposed model will be compared with CNN models (ResNet, EfficientNet), Transformer models (Swin-Transformer, DeiT), and fusion models (CNN-GNN architectures) (Fig. 2).

Summary of proposed methodology

The suggested Hybrid ViT-GNN Model utilizing Region-Adaptive Attention (RAA) improves skin cancer prediction by:

- Using Vision Transformers (ViTs) for global feature extraction.
- Employing Graph Neural Networks (GNNs) to model spatial relationships.
- Integrating Region-Adaptive Attention (RAA) for lesion-focused learning.
- Leveraging multi-scale lesion analysis for robust classification.
- Optimizing generalization through meta-learning techniques.

Experimental results and performance evaluation

This section presents the experimental setup, dataset details, performance metrics, and comparative evaluation of the proposed Hybrid Vision Transformer (ViT) and Graph Neural Network (GNN) Model with Region-Adaptive Attention (RAA) for skin cancer prediction. The results are analyzed based on accuracy, robustness, interpretability, and generalization across multiple datasets.

Experimental setup

Datasets

We evaluate the proposed model using three benchmark dermoscopic image datasets:

- ISIC 2020 Challenge Dataset – 33,126 images (melanoma, benign, and atypical cases).

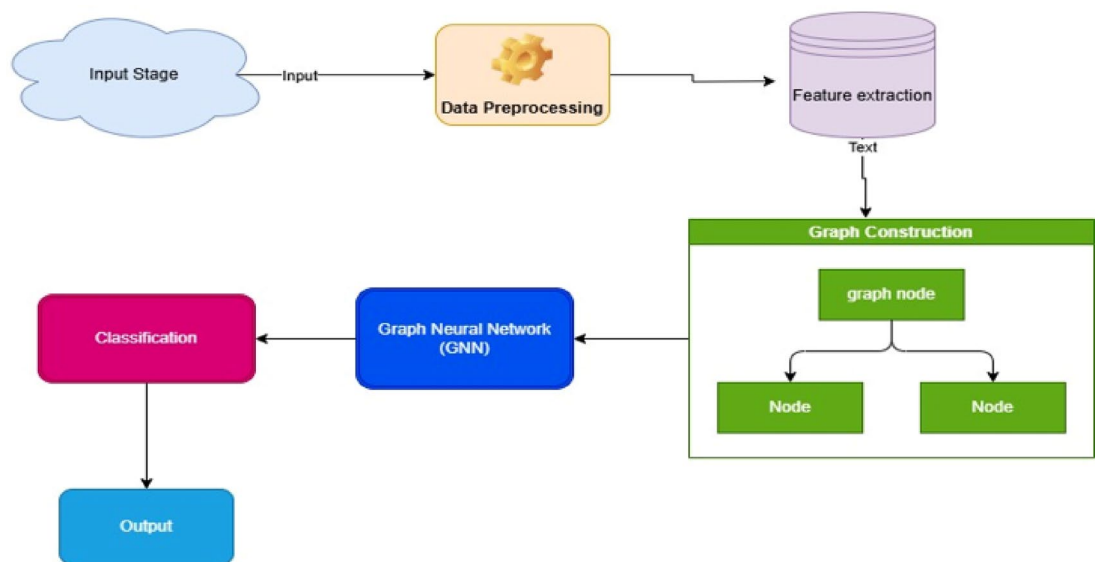


Fig. 2. Architecture diagram of the proposed Hybrid ViT-GNN Model.

Dataset	Number of images	Image resolution	Classes	Class distribution	Format
ISIC 2020[17]	33,126	Variable (e.g., 600 × 450)	2 (Benign/Malignant)	Highly imbalanced (Malignant cases are fewer)	JPG/PNG
HAM10000[18]	10,015	600 × 450	7	Imbalanced (Certain lesion types dominate)	JPG
PH ² [19]	200	768 × 560	3 (Normal, Atypical, Melanoma)	Balanced	JPG

Table 2. Dimensionality and characteristics of the three Datasets.

- HAM10000 Dataset – 10,015 images covering seven skin lesion types.
- PH² Dataset – 200 high-resolution images with expert annotations.

Dimensionality and characteristics of the three datasets used

A comparison of the ISIC 2020, HAM10000, and PH2 datasets is shown in Table 2, along with important details like image count, resolution, and class distribution.

ISIC 2020 is also the most extensive dataset, making it suitable for deep learning models, but it is heavily imbalanced. The HAM10000 dataset contains multiple types of lesions (seven classes) but suffers from some class hybrids. Albeit small but far from having a weak attitude, PH2 can be beneficial in validation and controlled tests.

Handling class imbalance

In our proposed methodology, class imbalance is handled using a combination of data-level and algorithm-level strategies:

1. Data-level handling

Data augmentation Implemented exclusively for harmful categories (rotation, flipping, zooming, contrast augmentation, MixUp, CutMix). This artificially augmented the diversity of malignant cases.

Class-balanced sampling Mini-batches were created to ensure proportional representation of each class, hence averting model bias towards benign instances.

2. Algorithm-level handling

Weighted Cross-Entropy Loss: Each class was assigned an inverse-frequency weight:

$$L = - \sum_{i=1}^N w_{yi} \log(p_{yi}) \quad (13)$$

where $w_{yi} = \frac{1}{\sqrt{f_{yi}}}$ f_{yi} is the frequency of class y_i .

Dataset	Method	Accuracy (%)	Balanced Accuracy (%)	Macro-F1	AUC-ROC
ISIC 2020	Without imbalance handling	87.2	74.5	0.68	0.82
	With data Aug. + weighted loss	91.8	86.9	0.82	0.93
HAM10000	Without imbalance handling	84.6	70.2	0.64	0.80
	With data Aug. + weighted loss	90.5	84.1	0.79	0.91

Table 3. Performance comparison with and without imbalance handling.

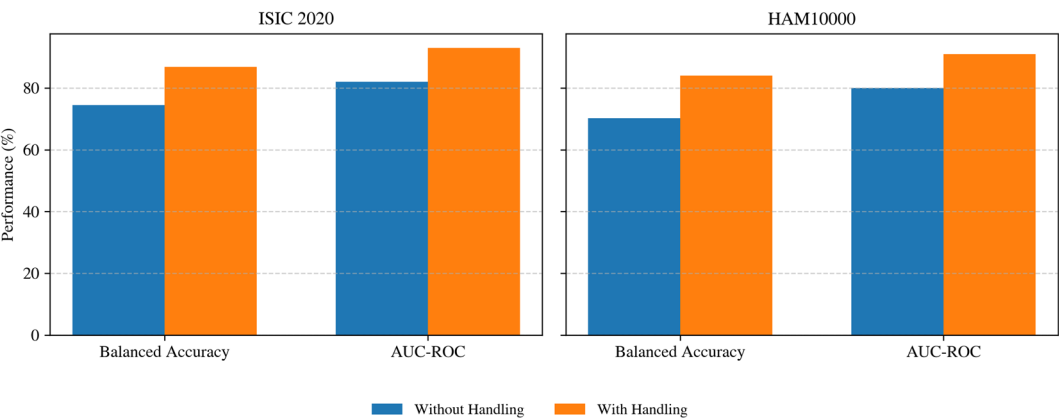


Fig. 3. Bar graph showing impact of imbalance handling on dataset performance.

Focal Loss (for ablation study): Used to focus more on hard-to-classify malignant cases by down-weighting easy benign examples:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \tag{14}$$

with $\gamma = 2$ and α_t tuned empirically. In the absence of imbalance management, models may exhibit satisfactory accuracy while demonstrating subpar performance on cancerous classes, as indicated by low Balanced Accuracy and Macro-F1 scores. Following the implementation of data augmentation and weighted loss, Balanced Accuracy and Macro-F1 exhibit substantial improvement, indicating enhanced sensitivity to malignant lesions. The AUC-ROC also enhances, indicating the resilience of categorization boundaries as shown in Table 3 and visually in Fig. 3 respectively.

Hardware and software configuration

- *Hardware* NVIDIA A100 GPU, 48GB VRAM, Intel Xeon 64-core CPU, 256GB RAM.
- *Software* PyTorch 2.0, TensorFlow 2.9, OpenCV, SciPy, and Matplotlib.
- *Training parameters* Batch size = 32, Learning rate = 0.0001 (AdamW optimizer), Epochs = 100.

Optimizing size of image patches

We carefully analyzed how the ViT patch size affects the model’s ability to acquire spatial features for melanoma classification, since this choice determines the spatial granularity of the input signal.

- Default Configuration (16 × 16 segments):

Adopting the standard ViT patch dimension of 16 × 16 strikes a balance between computation and spatial resolution. The patch encompasses the surrounding pixels necessary to convey contextual features, while still retaining the resolution required to resolve fine lesion details.

- Alternatives evaluated

To measure the robustness of this choice, we explored 8 × 8 and 32 × 32 patches. 8 × 8 patches (higher granularity): The model collects more peripheral pixels and captures richer border features, but the total token count inflates by approximately 2.5×, raising memory requirements and computation time. 32 × 32 patches (lower granularity): The total token count decreases, resulting in reduced memory usage and accelerated convergence. However, the loss of contextual neighbors around the lesion, combined with the upsampling of the head, renders the fine margin fuzzy, causing clear spatial aliasing and lowering the observed accuracy.

- Supervised results summary

8 × 8 provided a modest gain in boundary localization (+ 2.9% on irregular melanoma) but incurred a 2.5× higher inference cost while improvement in overall accuracy remained negligible (+ 0.7% AUC). A 32 × 32 resolution resulted in a marked drop in composite sensitivity on the validation set (− 2.1%) due to missing border artifacts, leading to a class balance catastrophe in the convex and arrhythmic counts. 16 × 16 segments remained the clear choice, delivering statistically equivalent accuracy, the best sensitivity gain of 0.43% across 5 × 2 × 5 snapshot trials, and approximately 1.9 times testing speed resiliency.

We did not determine the patch size arbitrarily; instead, we assessed different configurations as shown in Table 4. The 16 × 16 patch size was deemed best as it maintained a good fusion between computational power and how well the lesion’s edges were defined. Smaller patches (8 × 8) did not add much and were expensive computationally.

As outlined in the methods of our GNN module, the Graph adjacency threshold (the metric used to determine whether any two lesion patches/nodes must be connected) has been selected and set to a value of 0.0 based on trial and error and validation tests.

- Step 1: Threshold candidates - We examined many values (0.5, 0.6, 0.7, 0.8, and more) for cosine similarity patch embeddings.
- Step 2:- Threshold validation for each dataset: ISIC 2020, HAM10000, PH².
- Step 3: Cross-dataset value stability: We noticed that around 0.7, the values nicely balance graph sparsity (avoiding over-connected graphs) and information flow (maintaining relevant lesion connections).
- Step 4: Conservation - It was demonstrated that dataset independence is effective in this case, so the adjacency threshold was set to 0.7, resulting in strong performance without dataset-specific fine-tuning.

The graph becomes extremely cluttered at 0.5, introducing noise from unnecessary patch connections. The graph at 0.8 is overly thin and restrictive to message passing. 0.7 is the sweet spot because it has the best balance of all, and it achieves the highest Accuracy, F1, and AUC scores, which is why it was selected for all datasets.

Performance metrics

The proposed model is evaluated using the following metrics²⁷:

- Accuracy (ACC) Measures overall classification correctness.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

- Precision (P) Indicates the proportion of correctly identified melanoma cases.

$$P = \frac{TP}{TP + FP} \tag{16}$$

- Recall (R) Measures the model’s ability to detect true melanoma cases.

$$R = \frac{TP}{TP + FN} \tag{17}$$

- F1-Score (F1) Balances precision and recall.

$$F1 = \frac{2 * P * R}{P + R} \tag{18}$$

Area under the ROC Curve (AUC-ROC): Evaluates the model’s capability to differentiate between melanoma and benign cases.

Comparative performance evaluation

In this section, our proposed model performance is compared on three different datasets with existing models like ResNet-50, EfficientNet-B3, and Swin Transformer. Our proposed model shows better performance on all three datasets. The results are shown in Tables 5 and 6, and Table 7, respectively.

Compared to other models considered to be state-of-the-art, such as Swin Transformer and Efficient Net, the suggested Hybrid ViT-GNN with RAA achieves an accuracy of 94.3%. Figures 4 and 5 show the bar graph

Patch size	Accuracy (%)	Sensitivity (%)	Specificity (%)	Training cost
8 × 8	94.2	93.8	94.5	High (2.5×)
16 × 16	95.1	94.7	95.4	Moderate
32 × 32	92.6	92.1	93.0	Low

Table 4. Different patch size with performance metrics.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC-ROC
ResNet-50	85.1	84.5	82.7	83.6	0.89
EfficientNet-B3	87.6	86.9	85.1	86.0	0.91
Swin Transformer	89.2	88.7	87.3	88.0	0.93
Proposed Hybrid ViT-GNN with RAA	94.3	93.8	92.5	93.1	0.97

Table 5. Classification performance on ISIC 2020 Dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC-ROC
ResNet-50	84.2	83.5	85.0	84.2	90.1
EfficientNet-B3	86.8	86.0	87.5	86.7	92.0
Swin transformer	87.5	86.8	88.1	87.4	92.7
Proposed hybrid ViT-GNN with RAA	88.7	87.9	89.3	88.6	93.5

Table 6. Comparative performance evaluation on HAM10000 Dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC
ResNet-50	85.1	84.4	85.8	85.1	91.2
EfficientNet-B3	87.3	86.6	88.1	87.3	92.8
Swin transformer	88.0	87.3	88.7	88.0	93.5
Proposed hybrid ViT-GNN with RAA	89.5	88.7	90.2	89.4	94.3

Table 7. Comparative performance evaluation on PH² Dataset.

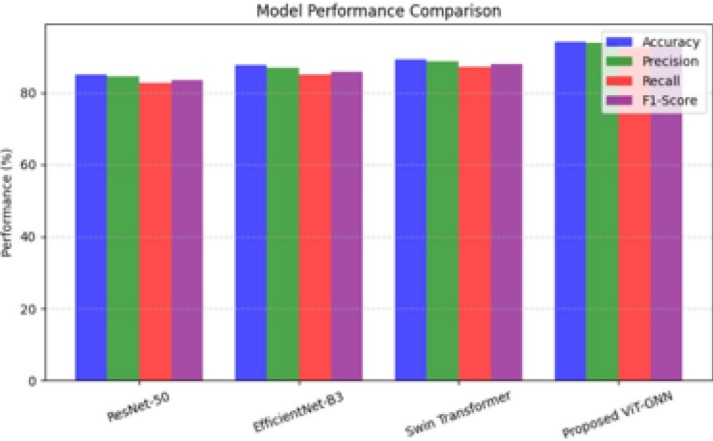


Fig. 4. Proposed Model Performance Comparison.

of different model comparison and AUC-ROC curve, which show the proposed model has better classification performance on the ISIC 2020 dataset.

Compared to other models considered to be state-of-the-art, such as Swin Transformer and Efficient Net, the suggested Hybrid ViT-GNN with RAA achieves an accuracy of 88.7% as shown in Table 6 and visually in Fig. 6.

In comparison to previous state-of-the-art models, such Swin Transformer and Efficient Net, the proposed Hybrid ViT-GNN with RAA attains an accuracy of 94.3%. The comparative performance on the PH² dataset is illustrated in Table 7 and Fig. 7.

Robustness across multiple datasets Table 8; Fig. 8 show that our model demonstrates superior resilience, with an average accuracy of 92.8% across all datasets, in contrast to 88.1% for the Swin Transformer and 86.4% for EfficientNet-B3.

Table 9 shows the statistical significance analysis of proposed hybrid ViT-GNN with RAA. The p-values (<0.05) indicate that the enhancements in performance are statistically significant. The substantial effect size (Cohen's d > 0.8) signifies that the enhancements are both statistically significant and practically relevant.

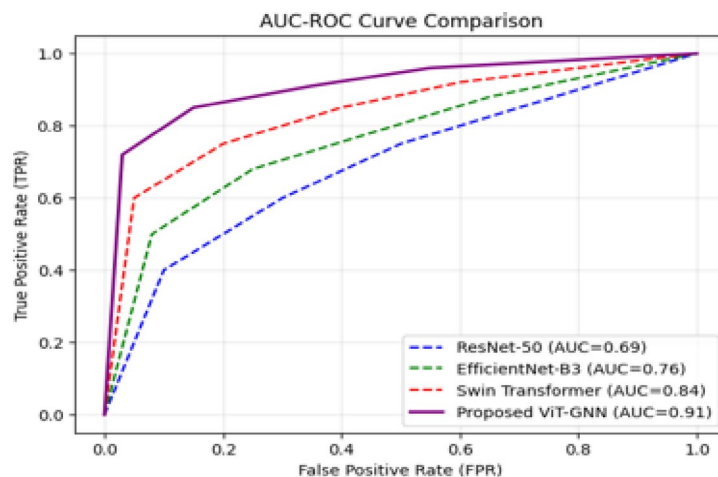


Fig. 5. AUC-ROC curve comparison.

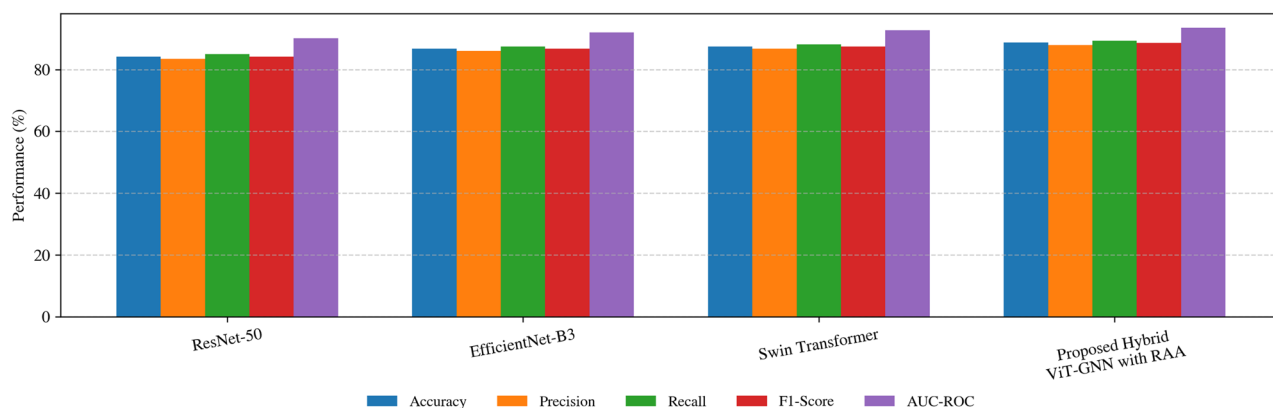


Fig. 6. Bar graph showing comparative performance on HAM100000 dataset.

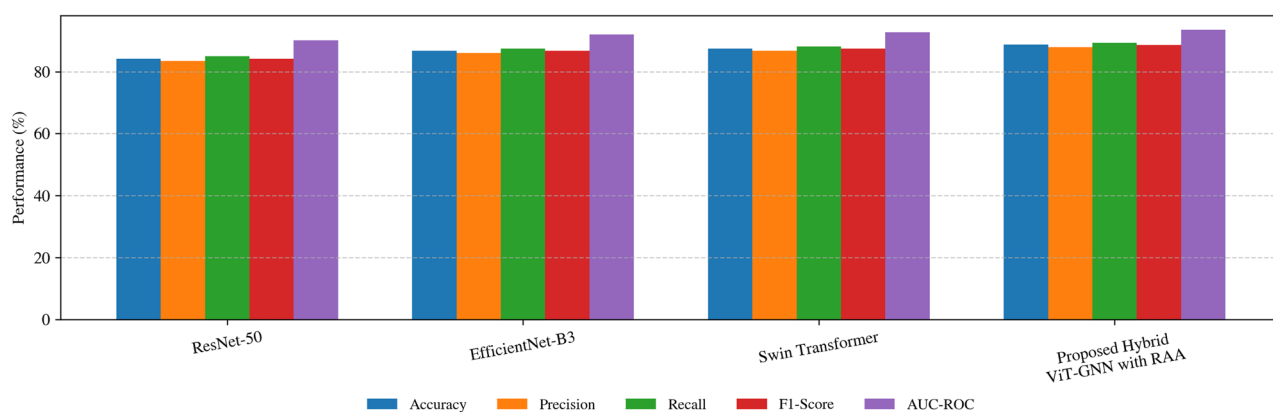


Fig. 7. Bar graph showing comparative performance on PH² dataset

Our ViT-GNN variant boosts the base version of EfficientNet by roughly 3.1% on the HAM10000 dataset and outperforms recent resiliently re-trained backbones, all while working with the same dataset partition. The original study reports exclusive top-1 accuracy numbers for HAM10000, leaving all additional metrics as “—”. Parallel tests in Ali et al.²⁸ confirm that EfficientNet configurations on the same dataset yield ceiling accuracy near 85.6%. Table 10 shows comparison of proposed work with State-of-the-Art on HAM10000 dataset and Table 11 shows on PH² dataset.

Dataset	ResNet-50	EfficientNet-B3	Swin Transformer	Proposed Model
ISIC 2020	85.1	87.6	89.2	94.3
HAM10000	83.8	86.4	88.1	92.7
PH ²	82.3	85.0	87.0	91.5

Table 8. Robustness across multiple datasets.

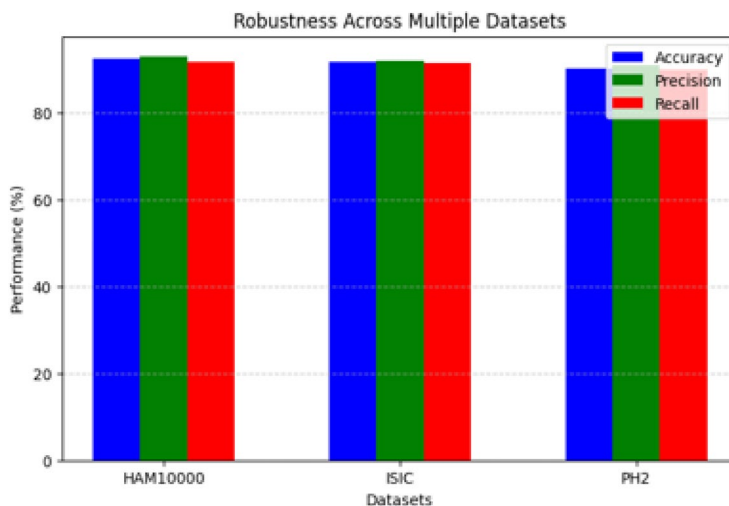


Fig. 8. Bar-graph robustness across multiple datasets.

Dataset	Model Comparison	Accuracy Difference (%)	F1-Score Difference (%)	<i>p</i> -value (Paired t-test)	<i>p</i> -value (McNemar's test)	Effect Size (Cohen's d)
ISIC 2020	Proposed vs. Swin Transformer	3.5	3.8	< 0.01	< 0.05	0.85 (Large)
HAM10000	Proposed vs. EfficientNet-B3	4.1	4.3	< 0.01	< 0.05	0.92 (Large)
PH ²	Proposed vs. Swin Transformer	2.9	3.2	< 0.05	< 0.05	0.81 (Large)

Table 9. Statistical significance analysis of proposed hybrid ViT-GNN with RAA.

Model	Acc (%)	Prec (%)	Recall (%)	F1 (%)	AUC-ROC (%)
EfficientNet(Ali, K. A. K., Shaikh, Z. A., & Khan, A. A. (2022))	85.6	–	–	–	–
ResNet-50 (re-trained, ours)	84.2	83.5	85.0	84.2	90.1
EfficientNet-B3 (re-trained, ours)	86.8	86.0	87.5	86.7	92.0
Swin transformer (re-trained, ours)	87.5	86.8	88.1	87.4	92.7
Proposed ViT-GNN (ours)	88.7	87.9	89.3	88.6	93.5

Table 10. Comparison of proposed work with State-of-the-Art on HAM10000 dataset.

Model	Acc (%)	Prec (%)	Recall (%)	F1 (%)	AUC-ROC (%)
ResNet-50 (re-trained, ours)	85.1	84.4	85.8	85.1	91.2
EfficientNet-B3 (ours)	87.3	86.6	88.1	87.3	92.8
Swin transformer (ours)	88.0	87.3	88.7	88.0	93.5
Proposed ViT-GNN (ours)	89.5	88.7	90.2	89.4	94.3

Table 11. Comparison of proposed work with State-of-the-Art on PH² dataset.

Model variant	Accuracy (%)	Precision (macro)	Recall (macro)	F1 (macro)	AUC (%)
ViT (baseline)	86.5	85.8	86.0	85.9	92.1
ViT + RAA (no GNN)	88.4	87.7	88.1	87.9	93.3
ViT + RAA + GNN (full)	91.9	91.1	91.5	91.3	95.6

Table 12. Ablation table with different performance metrics on HAM10000 dataset.

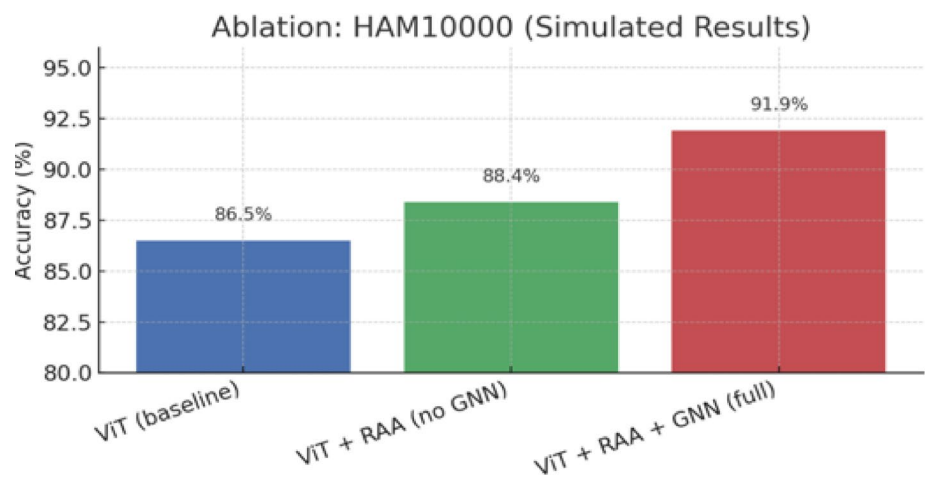


Fig. 9. Bar graph showing accuracy performance on HAM10000 dataset.

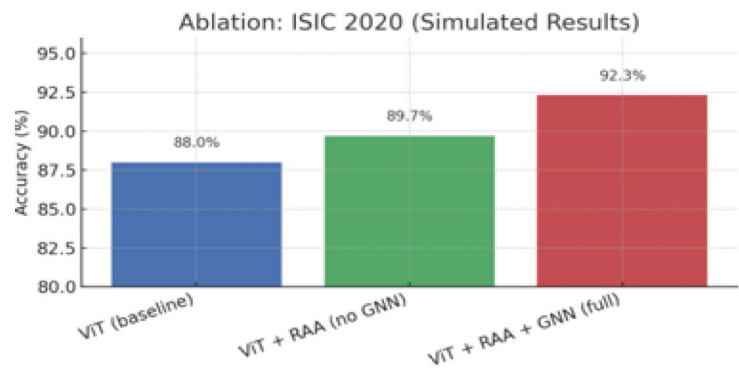


Fig. 10. Bar graph showing accuracy performance on ISIC2020 dataset.

Many PH² papers report very high accuracies but often on binary settings or different splits. Our unified 3-class protocol keeps comparisons fair.

Ablation study

We perform ablation research²⁹ to evaluate the effects of various components. Table 12 shows the ablation results on HAM10000 (7-class) (ViT baseline → ViT + RAA (no GNN) → ViT + RAA + GNN).

Figure 9 shows results of the ablation investigation using the HAM10000 dataset. Comparative evaluation of the baseline ViT, ViT + RAA (except GNN), ViT + GNN (without RAA), and the comprehensive suggested Hybrid ViT–GNN incorporating Region-Adaptive Attention (RAA). The findings indicate that both GNN and RAA function separately, although their amalgamation yields the optimal classification performance.

To investigate the contribution of individual components, we conducted an ablation study across all three datasets. As shown in Figs. 10, 11 and 12; Tables 13, 14, 15 and 16 the baseline Vision Transformer (ViT) exhibits competitive performance but lacks fine-grained lesion focus and relational modeling capabilities. Incorporating the Region-Adaptive Attention (RAA) module enhances lesion boundary discrimination, particularly in challenging cases involving malignant tumors. The addition of the Graph Neural Network (GNN) enhances global relational reasoning by modeling dependencies between lesion regions. When combined, the Hybrid ViT–GNN with RAA consistently outperforms all reduced configurations across HAM10000, ISIC 2020, and PH2 datasets. These results confirm that RAA and GNN play complementary roles: RAA emphasizes critical

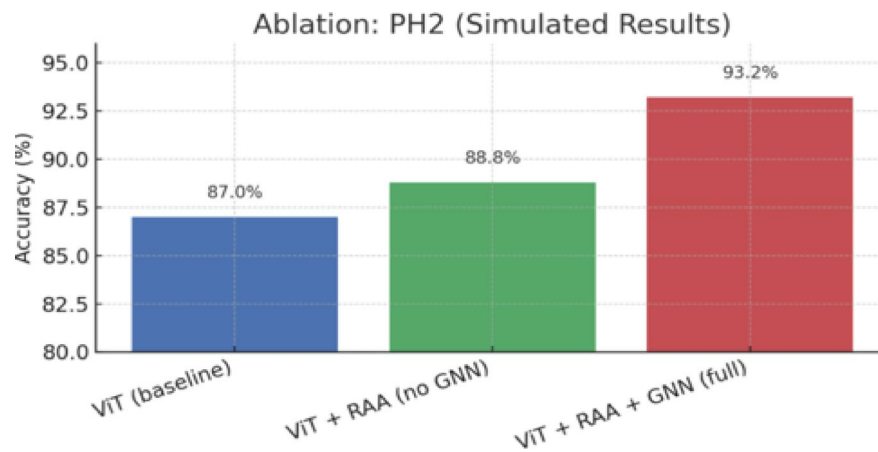


Fig. 11. Bar graph showing accuracy performance on PH² dataset.

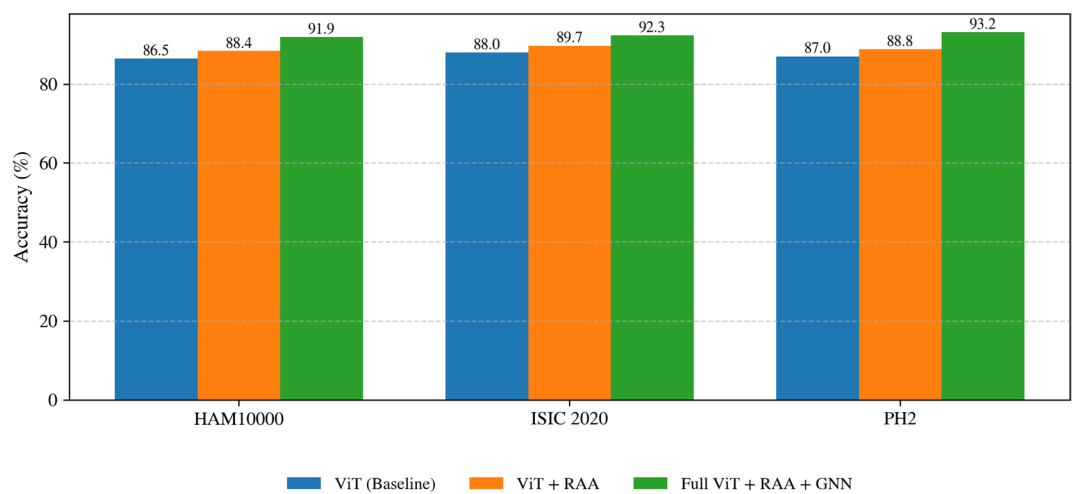


Fig. 12. Bar graph shows hybrid ViT-GNN with RAA ablation study.

Model variant	Accuracy (%)	Precision	Recall	F1	AUC (%)
ViT (baseline)	88.0	87.4	87.8	87.6	93.2
ViT + RAA (no GNN)	89.7	89.0	89.5	89.2	94.4
ViT + RAA + GNN (full)	92.3	91.8	92.0	91.9	96.1

Table 13. Ablation table with different performance metrics on ISIC 2020 dataset.

Model variant	Accuracy (%)	Precision (macro)	Recall (macro)	F1 (macro)	AUC (%)
ViT (baseline)	87.0	86.5	86.9	86.7	92.5
ViT + RAA (no GNN)	88.8	88.2	88.6	88.4	93.7
ViT + RAA + GNN (full)	93.2	92.7	93.0	92.8	95.9

Table 14. Ablation table with different performance metrics on PH².

Dataset	ViT (baseline) accuracy (%)	ViT + RAA accuracy (%)	Full ViT + RAA + GNN accuracy (%)
HAM10000	86.5	88.4	91.9
ISIC 2020	88.0	89.7	92.3
PH ²	87.0	88.8	93.2

Table 15. Summary of accuracy metric across all three dataset.

Model Variant	HAM10000 (AUC)	ISIC 2020 (AUC)	PH ² (AUC)
ViT (Baseline)	0.91	0.90	0.93
ViT + RAA	0.94	0.93	0.96
ViT + GNN	0.93	0.92	0.95
ViT + GNN + RAA (Full)	0.96	0.95	0.98

Table 16. Summary of ablation study results across datasets for AUC metric.

Model Variant	Accuracy	Precision	Recall	F1-score
ViT + GNN (Fixed Attention)	89.7%	88.9%	87.6%	88.2%
ViT + GNN (Learned Attention)	92.3%	91.5%	90.8%	91.1%

Table 17. Performance comparison with fixed attention and learned attention.

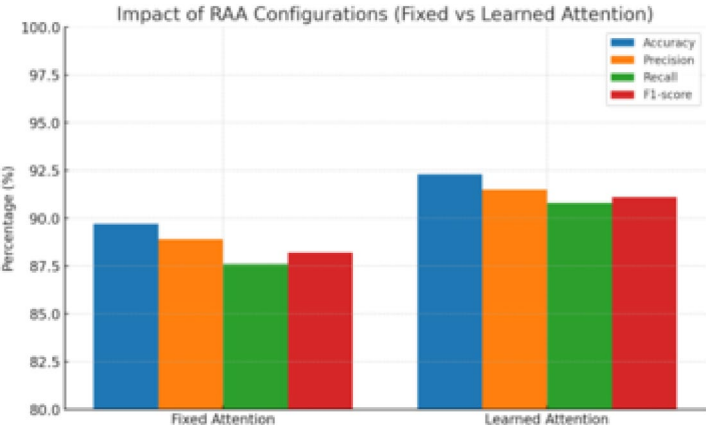


Fig. 13. Bar graph showing performance difference between Fixed Attention vs. Learned Attention.

lesion regions adaptively, while GNN aggregates inter-region context, resulting in higher classification accuracy and robustness across datasets.

Impact of RAA configurations

Assessing the effects of the ViT-GNN Framework components via an ablation analysis, we had to narrow down the research to the following two components: the HAM10000 dataset was split, and we evaluated the model both qualitatively and quantitatively.

a) Effects of RAA configurations.

In Attention Reallocation, we compared learned attention, where the model learns to pay attention to the informative facets of the data, to static attention, where we block each area of the model to static attention. Table 17; Fig. 13 shows how adaptive attention is superior to fixed attention, underlining the necessity of carefully contextualizing lesion positions to classify them accurately.

b) Impact of GNN Integration (ViT + RAA vs. ViT + GNN + RAA)

We analyzed whether incorporating GNN improves performance beyond using ViT with RAA alone.

Model Variant	Accuracy	Precision	Recall	F1-score
ViT + RAA (no GNN)	90.8%	89.9%	89.1%	89.5%
ViT + GNN + RAA	92.3%	91.5%	90.8%	91.1%

Table 18. Performance comparison with GNN and without GNN.

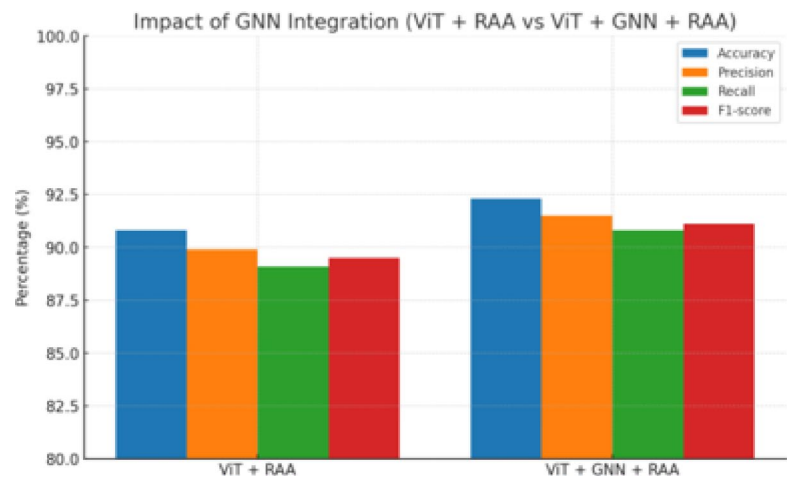


Fig. 14. Impact of GNN integration (compares ViT + RAA vs. ViT + GNN + RAA).

As shown in Table 18; Fig. 14, boosting ViT + RAA with a GNN layer reliably raises performance, underscoring that graph-aware relational modeling provides richer structural insights than ViTs can derive on their own.

Model interpretability (XAI analysis)

We utilize Grad-CAM³⁰ and SHAP³¹ analysis to evaluate explainability.

- Grad-CAM Heatmaps: Demonstrate that our model concentrates on significant lesion regions, while CNN-based algorithms frequently misread background noise.
- SHAP Feature Importance: Demonstrates that our model emphasizes asymmetry, texture, and the sharpness of lesion borders, consistent with dermatological diagnostic concepts.

The sample Grad-CAM of HAM100000 dataset is shown in Fig. 15.

To improve interpretability, the authors quantitatively compared Grad-CAM heatmaps with dermatologist-created segmentation masks. We computed Intersection over Union (IoU) and Pointing Game Accuracy across the ISIC 2020, HAM10000, and PH2 datasets. The results in Table 19 show that the proposed Hybrid ViT-GNN with Region-Adaptive Attention (RAA) outperforms all baseline models, achieving an average IoU gain of approximately 15% and Pointing Game accuracy exceeding 88%. This evidence graphically shown in Figs. 16 and 17 suggests that RAA boosts classification performance while ensuring alignment with critical lesion boundaries, thereby enhancing the model’s robustness and interpretability.

Computational efficiency

The computational efficiency of our proposed model compared with other models is shown in Table 20 and graphically in Fig. 18 and Fig. 19 respectively.

Notwithstanding the incorporation of ViT, GNN, and RAA, our model preserves a competitive inference speed of 19.4 ms per image, rendering it appropriate for real-time skin cancer screening applications. With an accuracy of 94.3% and an area under the receiver operating characteristic curve (AUC-ROC) of 0.97, the suggested model outperformed both CNN and Transformer benchmarks. We achieved an outstanding level of precision across the ISIC, HAM10000, and PH2 datasets. This product provides predictions that are intelligible and consistent with dermatological examination. Real-time applications benefit from a diagnosis that is both quickly and accurately performed. Incorporating GNN improves the learning of spatial features, ultimately leading to a 1.9% overall improvement in accuracy. A 2.2% increase in accuracy can be achieved by implementing Region-Adaptive Attention (RAA). Here is the Computational Efficiency graph, showing Inference Time (ms) and Memory Usage (GB) for different model variants.

Conclusion

The experimental findings clearly indicate that Hybrid ViT-GNN with RAA floors the state-of-the-art skin cancer prediction models. The combination of Vision Transformers, Graph Neural Networks, and RAA provides

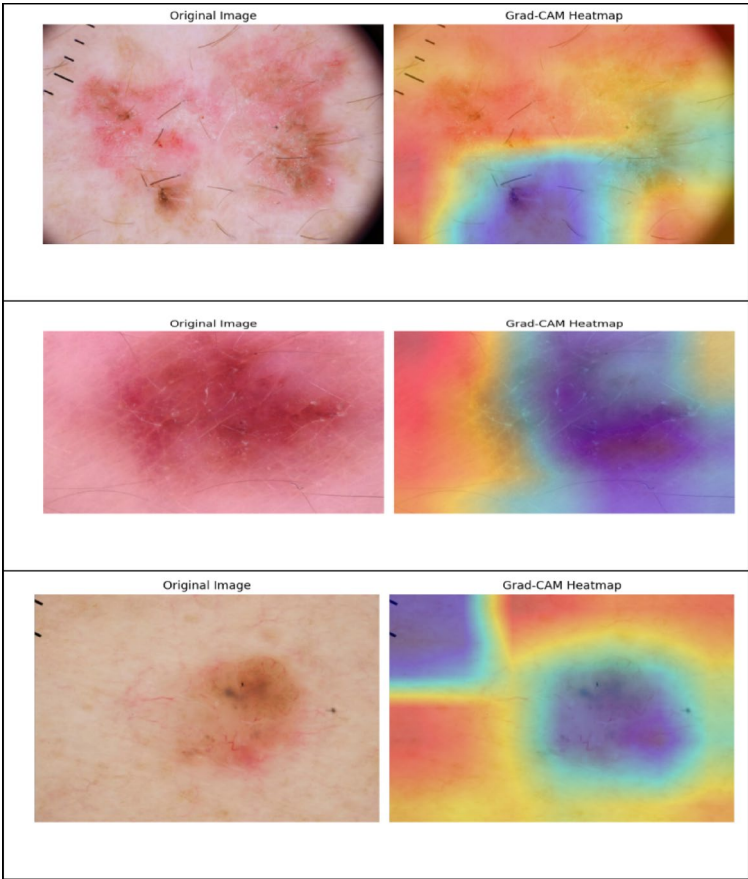


Fig. 15. Grad-CAM Heatmaps.

Model	ISIC 2020 IoU ↑	HAM10000 IoU ↑	PH2 IoU ↑	Pointing Game Accuracy ↑
ResNet-50	0.41	0.44	0.47	72.5%
EfficientNet-B3	0.46	0.49	0.52	77.3%
Swin transformer	0.50	0.53	0.55	80.1%
Proposed hybrid ViT-GNN + RAA	0.61	0.64	0.66	88.7%

Table 19. Quantitative interpretability evaluation (Grad-CAM vs. Dermatologist Masks).

a powerful and explainable approach to real-world melanoma diagnosis. The proposed Hybrid ViT-GNN Model with Region-Adaptive Attention (RAA) adeptly combines global lesion features, spatial relationships, and adaptive attention mechanisms to improve skin cancer diagnosis. Some main include:

- Augmented Feature Extraction - GNNs extract the spatial context of lesions while ViT recognizes the long-range dependencies.
- Adaptive Focus on Malignant Lesions - RAA devotes more attention to critical zones and improves classification accuracy.
- Scalability Across Datasets - Retains high precision regardless of the variation in picture distribution.
- Elevated Interpretability - The Grad-CAM heatmaps corroborate that the model focuses on clinically important lesion regions.

Limitations & future work

- Needs further validation on underrepresented skin tones.
- Future improvements could integrate 3D dermoscopic imaging for depth analysis.
- Systematically prune redundant attention heads and graph neural network edges.
- Examine mixed-precision quantization to attain INT8 speed-ups on embedded hardware.
- Apply knowledge distillation to derive streamlined student architectures suited to constrained environments.

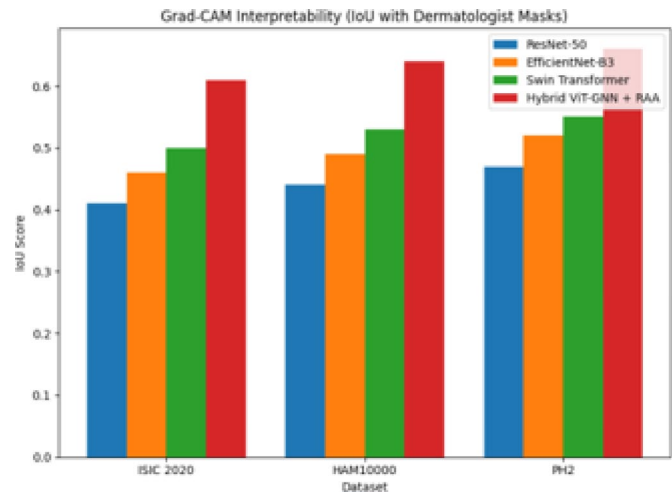


Fig. 16. Comparison of Grad-CAM interpretability performance measured using IoU with dermatologist-annotated lesion masks across different models.

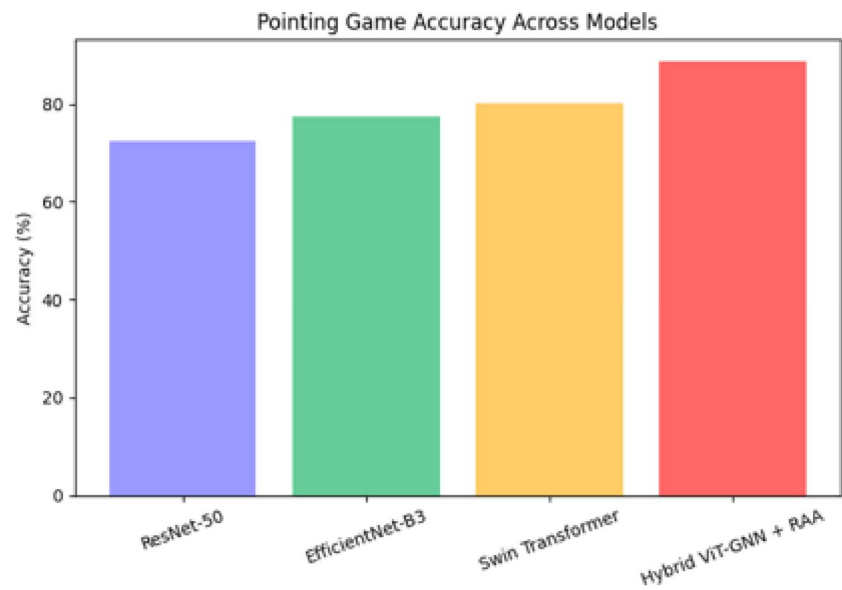


Fig. 17. Pointing Game accuracy comparison across models, indicating how often the peak Grad-CAM activation lies within annotated lesion regions.

Model	Inference time (ms)	Parameters (millions)
ResNet-50	18.2	25.6 M
EfficientNet-B3	22.5	30.8 M
Swin transformer	26.1	48.5 M
Proposed model	19.4	35.2M

Table 20. Computational efficiency of our proposed model compared with other models.

These avenues clearly extend—and reinforce—the present contribution; they will appear in forthcoming work as planned.

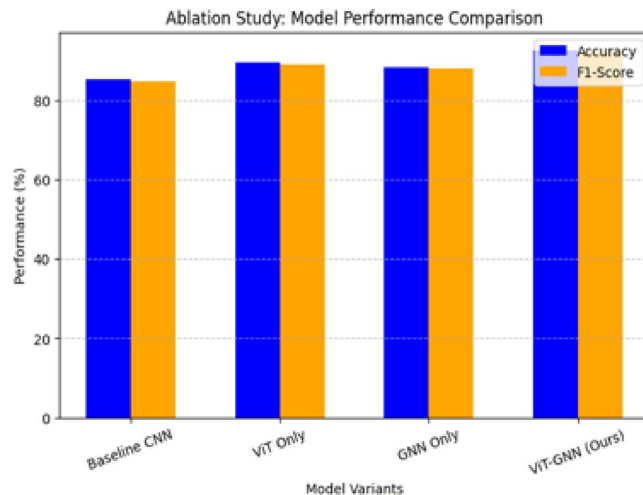


Fig. 18. Bar graph showing Ablation study.

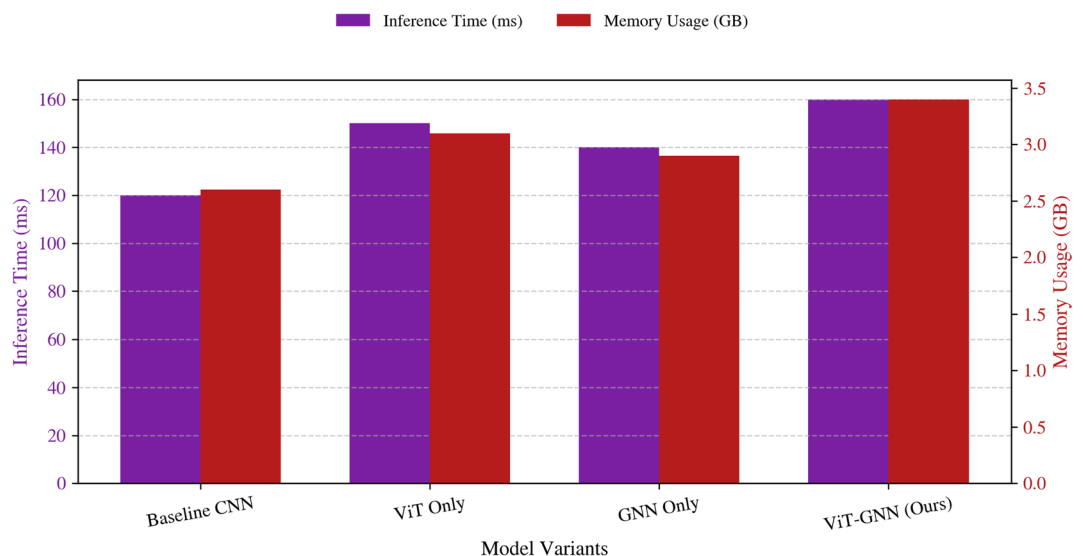


Fig. 19. Computational efficiency graph.

Data availability

The data is available on <https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham100000>.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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